

Simulation and Modeling: Discrete-Event System Simulation

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Fifth Edition 2014
Pearson Edition Limited 2014

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Syllabus

1. Basic concepts and terminology

- Concepts of a system
- System Methodology
- Advantages and disadvantages of simulation
- Simulation terminology

2. Probability and distribution theory

- Probability
- Set theory, compound events
- Conditional probability, independent events
- Discrete distributions
- Continuous distributions
- Function of a random variable
- Moments
- Some common distributions

3. Estimation and statistical tests

- Empirical distributions
- Estimation
- Tests of hypotheses
- The Chi-Square goodness-of-fit test
- The Kolmogorov-Smirnov test

4. Generation of Random Numbers

- Pseudo random numbers

- Congruential generators
- Testing and validating Pseudo random numbers

5. Introduction to queuing theory

- Review of the Poisson and Exponential distributions
- The M/M/1/∞/FIFO system
- Summary measures for the M/M/1/∞/FIFO system

6. Discrete system simulation

- Examples
- Time management methods
- Collecting and recording simulation data
- Analysis of simulation results
- Evaluation of the simulation model

7. Languages for discrete system simulation

- Language characteristics
- Use multipurpose languages
- Special-purpose languages: GPSS, SIMSCRIPT II.5, SLAM II, GASP IV

Grading Policy

- 4th week test: 10 Marks (Lecture)
- 7th week test: 15 Marks (Lecture)
- 7th week test: 5 Marks (Section)
- 12th week test: 20 Marks (Lecture)
- 5 assignments each one Mark (Lecture)
- 5 Marks Term work (Section)
- 40 Marks the Final Exam
- Any bonus during lectures will be added to the Final Marks

Simulation Principles and Methods

1. Introduction

The basic idea of Simulation is to build an experimental device (model) that acts like (simulates) the real system in certain important aspects. The purpose is to understand or evaluate the behavior of a complex real-world system over an extended period without risk to the real system performance.

2. Why Use Simulation?

a) **If experimentation on the real-world system is not feasible or not possible:**

- Too expensive (not feasible)
- Too risky (atomic problems) (not possible)
- Too complex
- Not existing system
- Intractable

b) **Simulation is the best (may be the only) technique available under conditions of uncertainty due to stochastic (random) variables or nonlinearity.** Mathematical treatment of such systems is frequently not possible.

c) **Computer simulation gives control over time**, which may be compressed or expanded, such as an aircraft; simulation may gather data on many months of operations in minutes.

3. Simulation Application

Simulation can be used in:

- Theoretical problems in basic science areas such as math, physics, and chemistry.
- Practical problems in all aspects of life such as industrial problems, business, economic, biomedical systems, social problems (Population dynamics), space research, and War strategies.

4. Basic Concepts and Terminology

a) **System:**

- Is defined as an isolated collection of interacting components or objects with a defined set of interactions among them.
- A Jet aircraft is an excellent example of a complex system consisting of mechanical, electronic, chemical, and human components.

b) **State of a system:**

- It is the minimal collection of information with which the system's future behavior can be predicted. For the Jet aircraft, the state of the system can be determined by Speed, Direction of travel, weather conditions, number of passengers, and amount of remaining fuel.

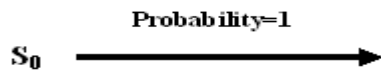


c) An Activity:

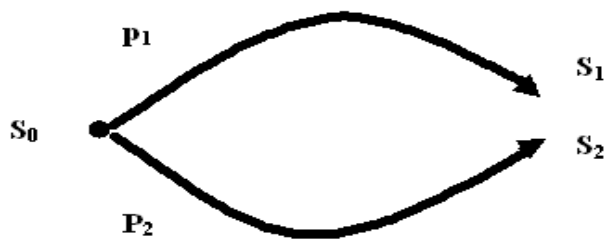
- It is the process or Event Which changes the system state.

d) System Classification:

- Natural and Man-Made
- Deterministic /



- Stochastic (Probabilistic)



- Continuous: A system whose changes in its state occur continuously over time (liquid flow in a pipeline)
- Discrete: Growth of world population with respect to time.

5. System Methodology

When simulation is used to solve a problem, the following time-tested steps, or stages are applied:

- Observation of the system
- Formulation of hypotheses or theories that account for the observed behavior
- Prediction of the future behavior of the system based on the assumption that the hypotheses are correct
- Comparison of the predicted behavior with the actual behavior

The system being studied may impose constraints on certain steps of this scientific method. For example, consider the simulation of a system that does not yet exist. Obviously, the observation of such a system is not possible, but the simulation of such a system may still be possible if the analysis is carefully conducted and if the ultimate requirements are known.

The problem-solving process is divided into four phases:

- Planning
- Modeling
- Validation
- Application

a) **Planning**

- Includes the initial encounter with the system, the problem to be solved, and the factors pertaining to the system and its environment that are likely to affect the solution of the problem. This means the problem must be well-defined. Obviously, the more accurate and precise the problem statement is, the more smoothly the solution process can proceed.
- Resources must be considered and estimated such as money, time, personnel, and special equipment. If crucial resources are not available, the solution of the problem can be judged infeasible before a significant amount of time or money is spent.

b) **Modeling**

- In this phase, the analyst constructs the system model, which is a representation of the real system. The characteristics of this model should be Representative (not identical) of the characteristics of the real system. You must select some minimal set of the system's characteristics so that the model approximates the real system to be cost-effective and manageable.
- There are many types of simulation models:
 1. Descriptive models (verbalizations of the system's composition)
 2. Physical models (scaled facsimiles of the system: aircraft, cars, building, etc.)
 3. Mathematical models (abstract expressions of the relationships among the system variables: Newton's laws of motion, etc.)
 4. Flowcharts (Logical interaction between system's components)
 5. Schematics (Logical interaction between system's components)
 6. Computer programs: If the system is too complex, then the reduction technique or subsystem modeling is used.

c) Validation

- Is to check that the system or the model is a correct representation of the real system. But verification means that the logic of a computer program is ok. So, a verified computer program can represent an invalid model.
- There are many techniques used to validate a computer program:
 - Compare the results of the simulation model with results historically produced by the real system operating under the same conditions.
 - Use a simulator to predict results, and then compare with the real data for some period.
 - Various statistical procedures can be used to determine whether the simulated output could have been produced by the real systems and vice versa. Two such methods are the Chi-square goodness-of-fit test and the Kolmogorov-Smirnov test.

d) Application

- Once the model has been properly validated, it can be applied to solve the problem at hand. A simulator may work error-free for a long period until it encounters a new and perhaps unique combination of program parameters that generates the next error.

6. Advantages and Disadvantages of Simulation

Phillips, Ravindran, and Solberg [OR principles and practice, New York: John Wiley, 1976] stated that simulation is one of the easiest tools of management science to use but probably one of the hardest to apply properly and perhaps the most difficult one from which to draw accurate conclusions.

Adkins and Pooch list 5 advantages of simulation modeling:

1. It permits controlled experimentation
2. It permits time compression
3. It permits sensitivity analysis by manipulation of input variables
4. It does not disturb the real system
5. It is an effective training tool

They also list 4 disadvantages:

1. A simulation model may become expensive in terms of manpower and computer time
2. Extensive development time may be encountered
3. Hidden critical assumption may cause the model to diverge from reality
4. Model parameters may be difficult to initialize. These may require extensive time in collection, analysis, and interpretation.

Thus, although simulation has proved an effective approach to problem-solving, it has its drawbacks. The researcher should be aware of these drawbacks before becoming committed to this approach.

Examples

Example 1

The first example we are going to see is the simulation of tossing a fair coin. The first step is analyzing the problem. The fair coin means that when tossing that coin, the probability of head equals the probability of tail equals 50%. So, using a digital computer to simulate this phenomenon, we are going to use a uniform Random number generated by the package you are using, or you can write its code. Uniform random number means that you have a set of random numbers between 0 and 1, all with the same probability. But most languages generate uniform random number integers from a to b with equal probability.

See the following program written in C++:

```
#include <iostream>
#include <stdlib.h>
void main(void)
{
    int x, number_of_trials, head = 0, tail = 0;
    float phead, ptail, error_head, error_tail;
    cout << "enter number of trials" << endl;
    cin >> number_of_trials;
    for (int i = 0; i < number_of_trials; i++)
    {
        x = random(2);
        if (x == 1) head++;
        else ++tail;
    }
    phead = head * 1.0 / number_of_trials;
    ptail = tail * 1.0 / number_of_trials;
    error_head = abs(((0.5 - phead) / 0.5) * 100);
    error_tail = abs(((0.5 - ptail) / 0.5) * 100);
    cout << "probability of head = " << phead << " with error =" <<
error_head << "%" << endl;
    cout << "probability of tail = " << ptail << " with error =" <<
error_tail << "%" << endl;
    cin >> x;
}
```

The output results are:

- Enter number of trials = 1
probability of head = 0 with error = 100%
Probability of tail = 1 with error = 100%
- Enter number of trials = 5
probability of head = 0 with error = 100%
Probability of tail = 1 with error = 100%
- Enter number of trials = 32767
probability of head = 0.496292 with error = 0%
Probability of tail = 0.503708 with error = 0%

Example 2

Get the average daily demand for a small grocery store selling fresh bread according to the following table:

Daily demand D	Probability Of demand P(D)	Device to generate chance outcomes Number and color of balls
100	0.20	20 Red
110	0.50	50 Blue
120	0.30	30 Yellow

Note that the proportion of balls of a specific color corresponds exactly to the probability of a specific level of daily demand.

To simulate the daily demand (5 days):

- Draw one ball at a time, notice its color, and then place it back in the bowl. Then translate the outcomes into unique values of demand.

Expected value of simulated demand = $560 / 5 = 112$ units/day

Analytical solution:

Expected daily demand = $100(0.2) + 110(0.5) + 120(0.3)$
= 111 units/day

Sample Number	Color of ball	Day of the Week	Simulated Demand
1	Blue	Monday	110
2	Blue	Tuesday	110
3	Yellow	Wednesday	120
4	Yellow	Thursday	120
5	Red	Friday	100
Total			560

Example 3

Use the one-digit Random Number (6, 3, 5, 0, 8) to generate random observations for:

- Throwing an unbiased coin
- Throwing a die
- The color of a traffic light found by a randomly arriving car when it is green 40% of the time, yellow 10% of the time, and red 50% of the time

Solution:

a) To simulate a coin using one digit R.N. (0-9), let 0–4 represent a Head and 5–9 represent a Tail; so the solution will be:

(6, 3, 5, 0, 8) → (T, H, T, H, T)

b) To simulate a die using numbers 0–9. Let 1–6 represent the faces of the die; then 0, 7, 8, and 9 are rejected. So the solution will be:

(6, 3, 5, 0, 8) → (6, 3, 5, reject, reject)

c) Divide 0–9 into 3 classes according to the given probability:

- Green 40%: 0, 1, 2, 3
- Yellow 10%: 4
- Red 50%: 5, 6, 7, 8, 9
So (6, 3, 5, 0, 8) → (Red, Green, Red, Green, Red)

Probability and Distribution Theory

Definition 1:

A random variable is a quantity whose value is determined by the outcome of a Random Experiment, i.e., the Random Variable X is a function whose domain is the event space (set of all possible outcomes of a random experiment) and whose range is some subset of the real numbers.

$$X: E \rightarrow R$$

Set Theory and Compound Events

Definition 2:

An event is some subset of the event space of a random experiment.

Ω : Event space or the collection of all the possible outcomes of the experiment.

Example:

Consider the experiment of tossing two dice and noting the sum of the faces. The event space is $\Omega = \{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$ and some events are:

$$E_1 = \{2\}$$

$$E_2 = \{3, 4, 5\}$$

$$E_3 = \{2, 4, 6, 8, 10, 12\} \text{ (an even number appears)}$$

Definition 3:

The Complement of an event E is $\bar{E}$, which contains the set of elements that are in Ω but not in E .

Note that: $E + \bar{E} = \Omega$, $E * \bar{E} = \Phi$ (null set)

Example:

$$\bar{E}_1 = \{3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$$

$$\bar{E}_2 = \{2, 6, 7, 8, 9, 10, 11, 12\}$$

$$\bar{E}_3 = \{3, 5, 7, 9, 11\}$$

Definition 4:

Let a random experiment have N equally likely outcomes. Let some event E be composed of n outcomes, so the probability of E is denoted by $P(E) = n/N$, $P(\bar{E}) = 1 - P(E) = (N - n)/N$

Note: $P(\Omega) = 1$, $P(\Phi) = 0$.

Definition 5:

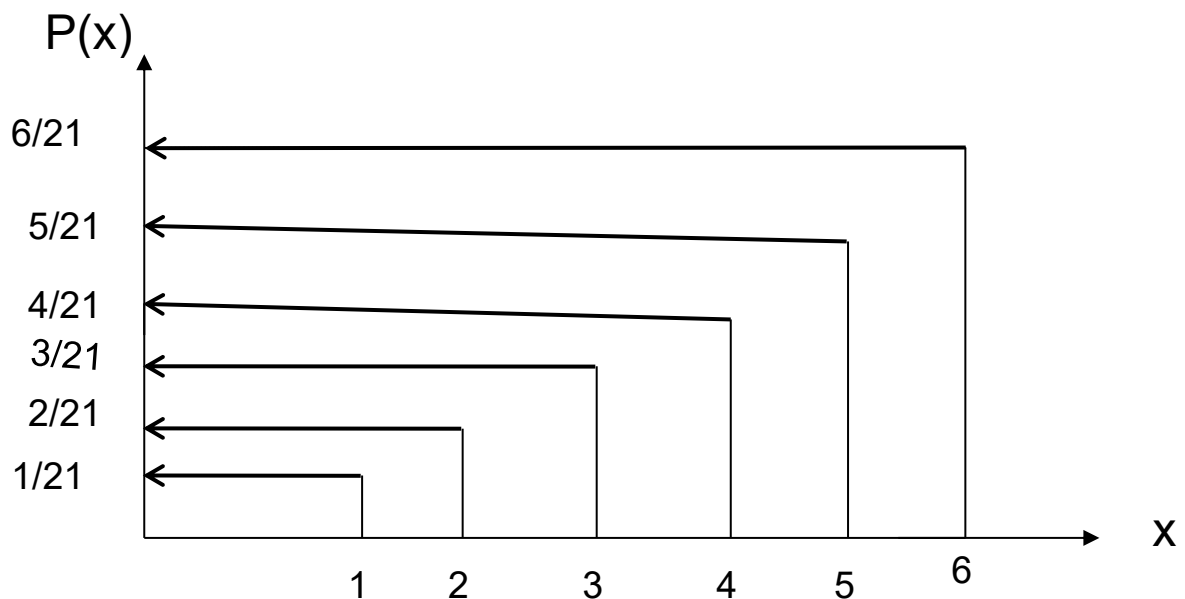
The Union of two events E_1 and E_2 is denoted by $E_1 + E_2$ and is defined as the outcomes in either E_1 OR E_2 OR both.

The intersection of two events E_1 & E_2 is denoted by $E_1 \cdot E_2$ and is defined as the outcomes that the events have in common. Two events that have no outcomes in common are said to be mutually exclusive.

Discrete Distribution

Example Tossing a single die Let X a random Variable that counts the spots of the side facing up. Then the values of X is 1, 2, 3, 4, 5, 6

X	1	2	3	4	5	6
$P(X = x)$	1/21	2/21	3/21	4/21	5/21	6/21



The properties that every discrete probability function must have:

1. $P(x_i) \geq 0$ for all x . It is called the probability density function (p.d.f.)
2. Another useful function is the Cumulative distribution $CDF = F(x) = P(X \leq x)$ {probability that $X \leq x$ }

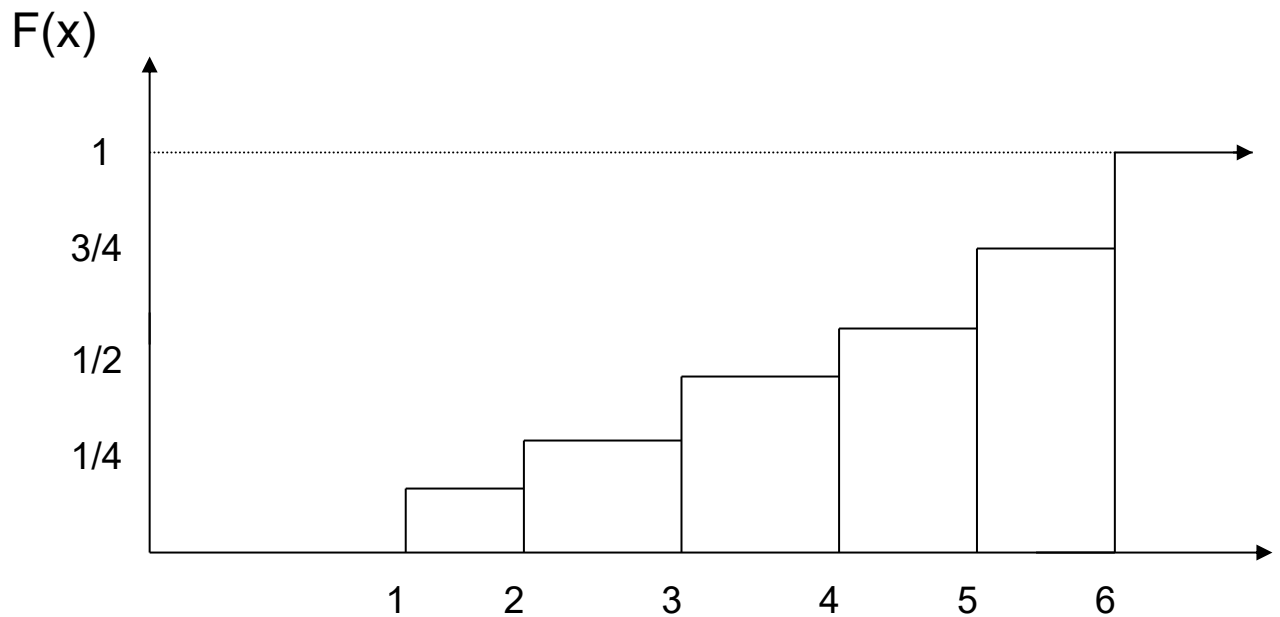
The properties that every discrete probability function must have :

- 1) $P(x_i) \geq 0$ for all x It is called probability density function (p.d.f.)

$$2) \sum_{i=1}^{\infty} P(X_i) = 1$$

Another useful function is the Cumulative distribution

$$\text{CDF} = F(x) = P(X \leq x) \quad \{\text{probability that } X \leq x\}$$



The CDF has the following properties:

- $0 \leq F(X) \leq 1, -\infty < x < \infty$
- If $X_1 \leq X_2$, then $F(X_1) \leq F(X_2)$; that is, F is monotonically increasing.
- $\lim_{x \rightarrow \infty} F(X) = F(\infty) = 1$ as $x \rightarrow \infty$
- $\lim_{x \rightarrow -\infty} F(X) = F(-\infty) = 0$ as $x \rightarrow -\infty$

Expected value (Mean):

$$E(X) = \sum x_i P(x_i)$$

For the above example:

$$E(X) = (1)(1/21) + (2)(2/21) + (3)(3/21) + (4)(4/21) + (5)(5/21) + (6)(6/21) = 91/21$$

Variance:

$$V(X) = \sum (x_i - \mu)^2 P(x_i)$$

For the above example, $V(X) = 2.222$

Standard Deviation of X is given by $S(X) = \sqrt{V(X)}$

For the above example, $S(X) = \sqrt{2.222} \approx 1.4907$

Continuous Distribution

1. $f(x) = 0$ if x is not in the range of X
2. $f(x) \geq 0$
3. $\int f(x) dx = 1$ over the range of X
4. $F(x_1) = \int_{-\infty}^{x_1} f(t) dt$
5. $E(X) = \int x f(x) dx$
6. $V(X) = E(X^2) - (E(X))^2 = \int (x - \mu)^2 f(x) dx$

Function of a Random Variable

Let X denote a R.V. Suppose the R.V. Y is related to X by $Y = aX + b$ where a, b are arbitrary constants.

$$E(Y) = E(aX + b) = aE(X) + b$$

$$\int_{-\infty}^{\infty} (ax + b) f(x) dx = a \int_{-\infty}^{\infty} xf(x) dx + b \int_{-\infty}^{\infty} f(x) dx$$

$$V(Y) = V(aX + b) = a^2 V(X)$$

$$= \int_{-\infty}^{\infty} (ax + b)^2 f(x) dx - (E(aX + b))^2$$

$$= \int_{-\infty}^{\infty} a^2 x^2 f(x) dx + 2ab \int_{-\infty}^{\infty} xf(x) dx + b^2 \int_{-\infty}^{\infty} f(x) dx - (E(aX + b))^2$$

$$= \int_{-\infty}^{\infty} a^2 x^2 f(x) dx + 2abE(X) + b^2 - (E(aX + b))^2$$

$$= \int_{-\infty}^{\infty} a^2 x^2 f(x) dx - a^2 (E(X))^2$$

$$= a^2 \left[\int_{-\infty}^{\infty} x^2 f(x) dx - (E(X))^2 \right]$$

$$= a^2 V(X)$$

Some Common Distributions

1. Bernoulli Distribution

It is a discrete distribution having only two outcomes. Let X be defined such that it takes the value 0 with probability p and 1 with probability $q = 1 - p$, i.e.:

$X \mid 0 \mid 1$

$P(x) \mid p \mid q$

CDF = $F(x) =$ **Example:**

Consider the experiment of tossing a fair coin. Let X assume the value 0 if a head appears and 1 for tail. Then $p = q = \frac{1}{2}$ and μ (mean) = $q = \frac{1}{2}$.

- 0 for $x < 0$

- p for $0 \leq x < 1$

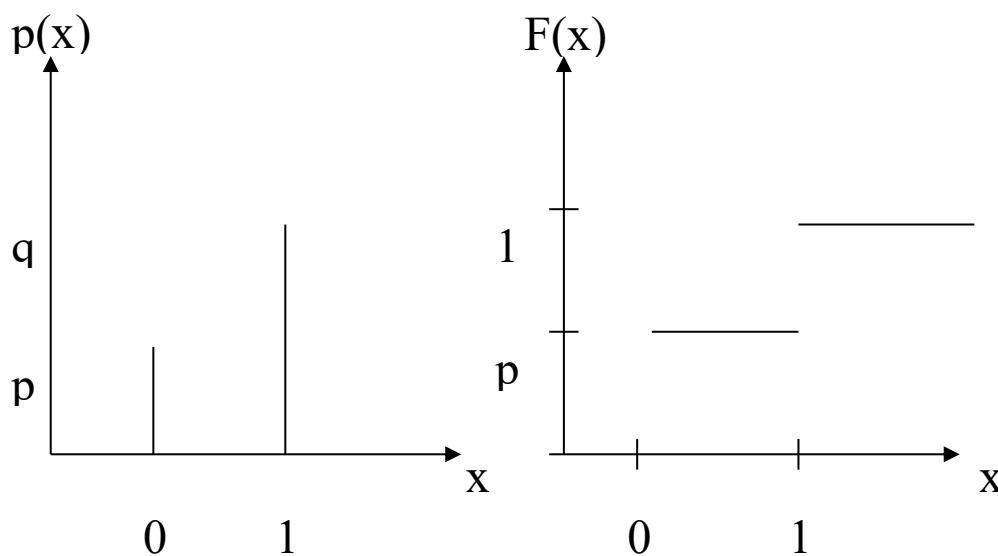
- 1 for $x \geq 1$

$$\mu = E(X) = 0 \cdot p + 1 \cdot q = q$$

$$\sigma^2 = V(X) = E(X^2) - (E(X))^2$$

$$= 0 \cdot p + 1 \cdot q - q^2 = q(1 - q) = pq$$

- $\sigma = (pq)^{1/2}$



Example

Consider the experiment of tossing a fair coin. Let X assume the value 0 if a head appears and 1 for tail. Then $p = q = \frac{1}{2}$ and μ (mean) = $p = \frac{1}{2}$ &

- $\sigma^2 (V(X)) = pq = 1/4$

2. Uniform Distribution

Let variable X have a uniform distribution between a and b : $X \approx U(a, b)$

$f(x) = 1/(b - a)$ for $a \leq x \leq b$, 0 otherwise

The interval $[a, b]$ is called the range of the distribution.

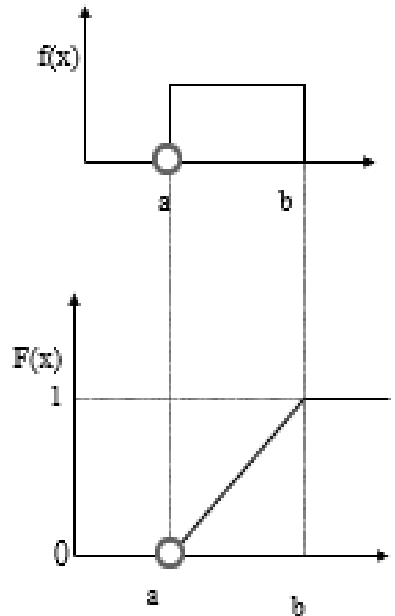
CDF is given by: $F(x) = (x - a)/(b - a)$ for $a \leq x \leq b$

$$F(X) = \int_{-\infty}^x f(t) dt = 0 \quad x < a$$

$$= \frac{x - a}{b - a} \quad a \leq x \leq b$$

$$= 1 \quad x > b$$

$$\text{Mean} = \mu = \frac{a + b}{2} \quad \text{Var.} = \sigma^2 = \frac{(b - a)^2}{12}$$



3. Poisson Distribution

It is a discrete distribution.

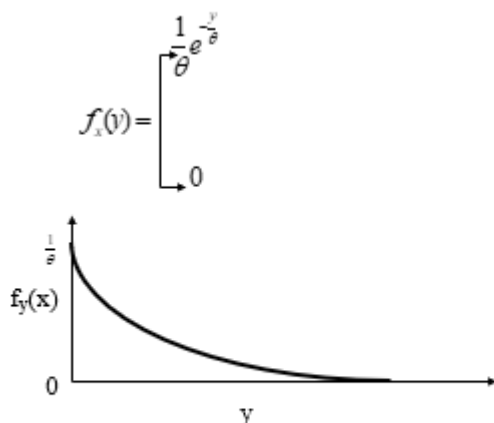
$P(X = k) = (\lambda^k e^{-\lambda}) / k!$

Mean = λ , Variance = λ

4. Exponential Distribution

$f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$

$F(x) = 1 - e^{-\lambda x}$



$$\int_{-\infty}^{\infty} f_x(y) dy = \int_0^{\infty} \frac{1}{\theta} e^{-\frac{y}{\theta}} dy = -e^{-\frac{y}{\theta}} \Big|_0^{\infty} = 1$$

$$\mu = E(X) = \frac{1}{\theta} \quad , \quad \sigma^2 = V(X) = \frac{1}{\theta^2}$$

$$CDF = F_x(b) = \int_{-\infty}^b f_x(y) dy = 0 \quad b < 0$$

$$= \int_0^b \frac{1}{\theta} e^{-\frac{y}{\theta}} dy = 1 - e^{-\frac{b}{\theta}} \quad b \geq 0$$

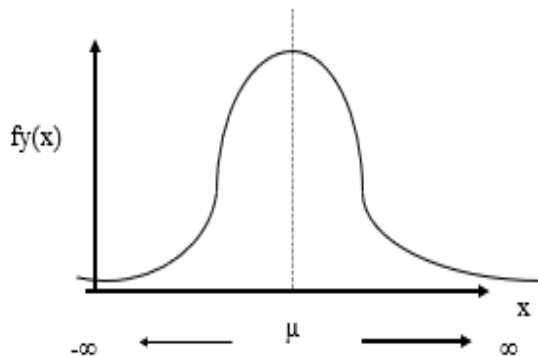
5. Normal Distribution

$$f(x) = (1/(\sigma\sqrt{2\pi})) e^{-(x-\mu)^2/(2\sigma^2)}$$

This integral does not exist in closed form. But there exist tables of values for the standard normal distribution with mean = 0 and var. = 1, which is given by: $Z = (X - \mu)/\sigma$

$$f_y(y) = \frac{1}{\sqrt{2\pi}\sigma} e^{\frac{-(y-\mu)^2}{2\sigma^2}} \quad -\infty < y < \infty$$

it is called a $N \approx (\mu, \sigma)$ i.e., it has two parameters.



$$F_x(b) = \int_{-\infty}^b \frac{1}{\sqrt{2\pi}\sigma} e^{\frac{-(y-\mu)^2}{2\sigma^2}} dy$$

This integral does not exist in closed form. But there exist tables of values for the standard normal distribution with mean=0 and var.=1 which is given by:

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{\frac{-z^2}{2}} \quad (\mu=0, \sigma=1)$$

If X is $N \approx (\mu, \sigma)$ then the random variable Z is

$$Z = (X - \mu) / \sigma \text{ is a } N \approx (0,1)$$

$\therefore X = \mu + Z\sigma$ i.e., X is $N \approx (\mu, \sigma)$

Variance Reduction Techniques

1. Increase sample size N where $V = E(x^2) - (E(X))^2$, $V \propto 1/N$

Examples:

Use $f(x) = e^{-x}$, an exponential distribution with mean $\lambda = 1$, i.e., $F(X) = 1 - e^{-x}$.

Let $X_i = -\ln(1 - r_i)$

i	r_i	$X_i = -\ln(1 - r_i)$
1	0.495	0.684
2	0.335	0.408
3	0.791	1.568
4	0.469	0.633
5	0.279	0.326
6	0.698	1.199
7	0.013	0.014
8	0.761	1.433
9	0.290	0.343
10	0.693	1.183

Total = 7.793, so mean = $7.793/10 = 0.779$ with error = $((1 - 0.779)/1)*100 = 22.1\%$

2. Use Complement Random Numbers for the above example $f(x) = e^{(-x)}$ i | ri | Xi = -

i	ri	$X_i = -\ln(1 - r_i)$	$\bar{r}_i = 1 - r_i$	$\bar{X}_i = -\ln(1 - \bar{r}_i)$
1	0.495	0.684	0.505	0.702
2	0.335	0.408	0.665	1.092
3	0.791	1.568	0.209	0.234
4	0.469	0.633	0.531	0.756
5	0.279	0.326	0.721	1.275
6	0.698	1.199	0.302	0.359
7	0.013	0.014	0.987	4.305
8	0.761	1.433	0.239	0.272
9	0.290	0.343	0.710	1.236
10	0.693	1.183	0.307	0.366
Total		7.793		10.597
Estimated mean	$0.5(0.7793 + 1.0597) = 0.92$			
With Error	$((1 - 0.92)/1) * 100 = 8.05\%$			

Example:

Determine the mean of the random variable with $F(X) = x^2$, $0 \leq x \leq 1$; $f(x) = 2x$, $0 \leq x \leq 1$.

$$\bar{x} = \int_0^1 x f(x) dx = \int_0^1 2x^2 dx = 2/3$$

$F(X) = X^2 = r$, so $X_i = \sqrt{r_i}$

ri	$X_i = \sqrt{r_i}$
0.628	0.792
0.207	0.455
0.662	0.814
0.774	0.880
0.360	0.600
0.951	0.975

TOTAL = 4.516, Estimated mean = $4.516/6 = 0.753$ with error = 13%

Using Complement Random Variables:

r_i	$\bar{r}_i = 1 - r_i$	$x_i = \sqrt{r_i}$	$\sqrt{(1 - r_i)}$
0.628		0.792	
	0.372		0.61
0.207		0.455	
	0.793		0.891
0.662		0.814	
	0.338		0.581

Total = $2.061 + 2.082 = 4.143$

So the mean = $4.143/6 = 0.691$ with Error = $((2/3 - 0.691)/(2/3)) * 100 = 4\%$

Introduction to Queuing Theory

A queuing system is a system in which customers arrive, wait if that service is not immediately available, receive the necessary service, and then depart.

Customers arriving $\rightarrow$ Service facility $\rightarrow$ Customers departing

Characteristics of any queuing system

- (A) The arrival pattern (distribution of arriving customers)
- (B) The service process (distribution of service time)
- (C) The number of parallel servers
- (D) The system capacity
- (E) The queuing discipline (FIFO, LIFO, priority)

So any queuing system can be described as A/B/C/D/E

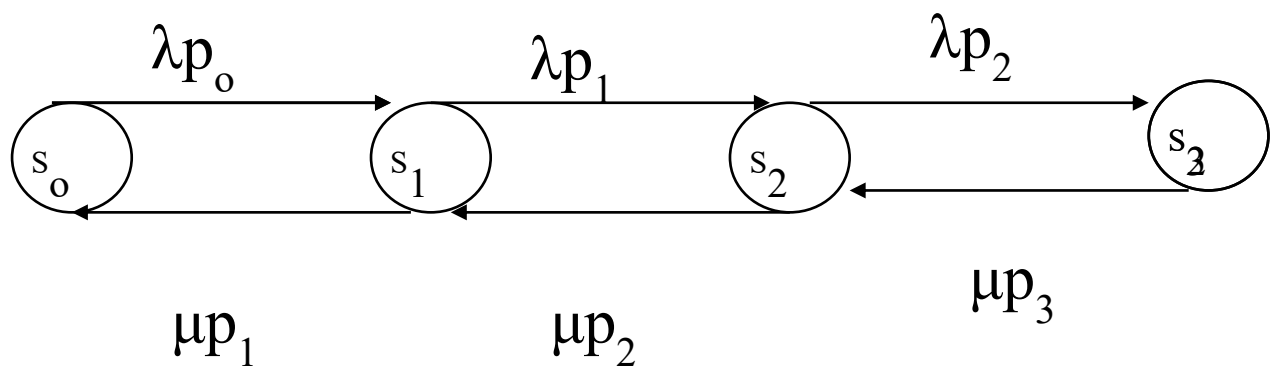
The items of interest concerning any queuing model:

1. Queuing length (L_q)
2. Time in the system (W)
3. Idle and busy time of a server

M/M/1/ ∞ /FIFO system

Derivation using stochastic balance:

Using balance at point S_0 : $\mu p_1 = \lambda p_0 \Rightarrow p_1 = (\lambda/\mu) p_0$



At point S_1 : $(\lambda + \mu)p_1 = \lambda p_0 + \mu p_2 \Rightarrow p_2 = (\lambda/\mu)^2 p_0$

And so on, we get $p_n = (\lambda/\mu)^n p_0$

$$\sum_{n=0}^{\infty} p_n = 1 = \sum_{n=0}^{\infty} \left(\frac{\lambda}{\mu}\right)^n p_0$$

$$\therefore p_0 = \frac{1}{\sum_{n=0}^{\infty} \left(\frac{\lambda}{\mu}\right)^n} = \frac{1}{\sum_{n=0}^{\infty} \rho^n} =$$

$$\therefore p_0 = \frac{1}{1 - \rho}$$

$$\therefore p_n = (1 - \rho) \rho^n \quad \text{for } n = 0, 1, 2, 3, \dots$$

$$L = \frac{\rho}{1 - \rho} \quad \rho < 1,$$

$$L_q = \frac{\rho^2}{1 - \rho} \quad \rho < 1$$

Expected number in the system (L): $L = E(X) = \sum n p_n$

$$= \sum_{x=0}^{\infty} x p_n = \sum_{x=0}^{\infty} x (1 - \rho) \rho^x$$

$$L = (1 - \rho) \sum_{x=0}^{\infty} x \rho^x$$

$$= (1 - \rho) \rho \sum_{x=0}^{\infty} x \rho^{x-1}$$

$$= (1 - \rho) \rho \frac{d}{d\rho} \sum_{x=0}^{\infty} \rho^x$$

$$= (1 - \rho) \rho \frac{d}{d\rho} \frac{1}{1 - \rho}$$

$$= (1 - \rho) \rho \frac{1}{(1 - \rho)^2}$$

$$\therefore L = \frac{\rho}{1 - \rho} = \frac{\lambda}{\mu - \lambda} \quad \rho < 1 \text{ i.e., } \mu > \lambda$$

Expected Number in the queue $L_q = \sum (n - 1) p_n$

$$L_q = 0p_0 + \sum_{x=1}^{\infty} (x-1)p_x$$

$$= \sum_{x=1}^{\infty} xp_x - \sum_{x=1}^{\infty} p_x$$

$$= L - (1 - p_0)$$

$$= L - 1 + p_0$$

$$= \frac{\rho}{1-\rho} - 1 + 1 - \rho$$

$$\therefore L_q = \frac{\rho^2}{1-\rho} = \frac{\lambda^2}{\mu(\mu-\lambda)} \quad \rho < 1, \text{ i.e., } \mu > \lambda$$

Using Little's formula:

The Expected time in the system $(W) = L / \lambda$

$$W = \frac{L}{\lambda}$$

$$= \frac{1}{\mu - \lambda} \quad \mu > \lambda$$

The Expected time in the Queue $(w_q) = L_q / \lambda$

$$W_q = W - \frac{1}{\mu}$$

$$= \frac{\lambda}{\mu(\mu - \lambda)} \quad \mu > \lambda$$

Random Number Generation (Pseudo R.N.)

There are many algorithms for random number generations, but we are going to emphasize the most common one, which is known as the Congruential method.

$R_{i+1} = (a R_i + c) \text{ modulo } T$
where:

- R_{i+1} is the new uniform random number
- R_i is the previous uniform random number
- a , c , and T are arbitrary constants

Example:

For $a = 1$, $c = 4$, $T = 5$, $R_0 = 2$ (seed); generate 10 random numbers.

Solution:

$R_0 = 2$

$$R_1 = \text{Rem}[\{(1)(2) + 4\}/5] = 1$$

$$R_2 = \text{Rem}[\{(1)(1) + 4\}/5] = 0$$

$$R_3 = \text{Rem}[\{(1)(0) + 4\}/5] = 4$$

$$R_4 = \text{Rem}[\{(1)(4) + 4\}/5] = 3$$

$$R_5 = \text{Rem}[\{(1)(3) + 4\}/5] = 2$$

$$R_6 = \text{Rem}[\{(1)(2) + 4\}/5] = 1$$

$$R_7 = \text{Rem}[\{(1)(1) + 4\}/5] = 0$$

$$R_8 = \text{Rem}[\{(1)(0) + 4\}/5] = 4$$

$$R_9 = \text{Rem}[\{(1)(4) + 4\}/5] = 3$$

$$R_{10} = \text{Rem}[\{(1)(3) + 4\}/5] = 2$$

Generation of a Random Observation Following a Certain Distribution

1. **For a Simple Discrete Distribution**, we use the allocation method (see the previous examples).
2. **For Continuous Distribution**, we use the inverse method if and only if the distribution is reversible (see the following examples).

Example 1:

Generate a random observation from the uniform distribution between a and b , i.e., $U(a, b)$.

$$f(x) = \frac{1}{b-a}$$

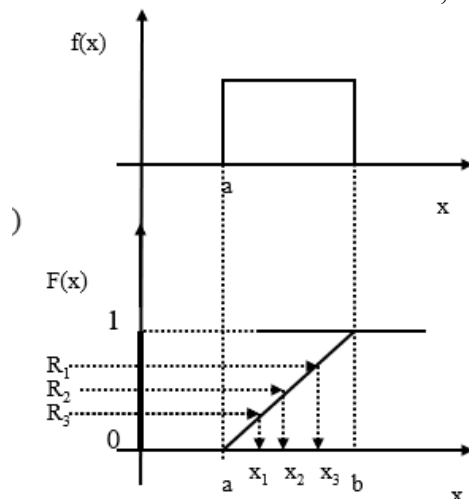
$$F(x) = \frac{x-a}{b-a}$$

$$U(0,1) \longrightarrow U(a,b)$$

$$R = F(x) = \frac{x-a}{b-a}$$

$$\therefore x = a + R(b-a)$$

$$\text{i.e., } x = F^{-1}(R)$$



Most Important Example:

Generate Random Observations from Exponential distribution with mean μ .

$$f(x) = \mu e^{-\mu x}$$

$$F(x) = 1 - e^{-\mu x}$$

$$R = F(x) = 1 - e^{-\mu x}$$

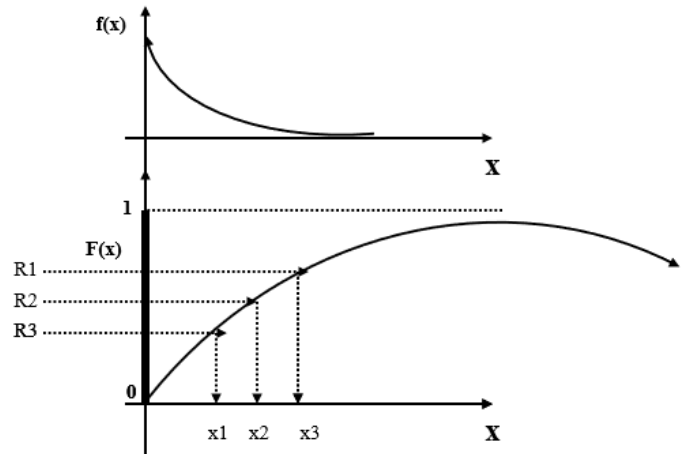
$$\therefore x = \frac{\ln(1 - R)}{-\mu}$$

$$\text{so } x_1 = \frac{\ln(1 - R_1)}{-\mu}$$

$$x_2 = \frac{\ln(1 - R_2)}{-\mu}$$

.....

where $R_1, R_2, \dots$, are $U(0,1)$



Illustrative Example: Sales of Life Insurance Problem

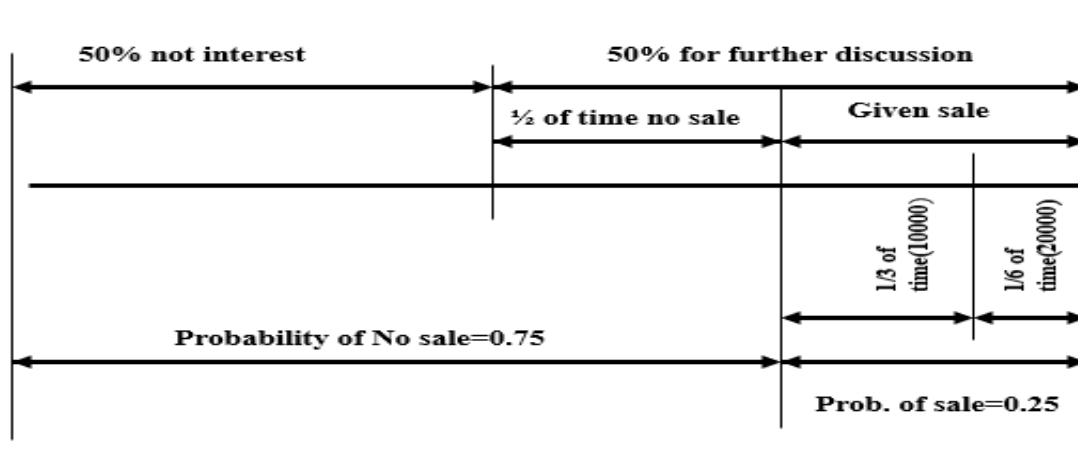
From the previous sales history, each time a house call is made:

- 50% not interested in purchasing
- 50% willing for further discussions

After discussions:

- $\frac{1}{2}$ of the time the visit will result in “No Sale”
- $\frac{1}{3}$ of the time will result in a sale of a 10,000 L.E. policy
- $\frac{1}{6}$ of the time will result in a sale of a 20,000 L.E. policy

Using a simulated sample of 20 visits, determine the probability that a sale is made and the expected policy value for each policy, assuming a sale is made.



Theoretical Solution:

- Probability of No Sale = 0.75
- Probability of Sale = 0.25
- Probability(10000 policy | sale) = $2/3 \approx 0.667$
- Probability(20000 policy | sale) = $1/3 \approx 0.333$

Expected value = $0.667 \times 10000 + 0.333 \times 20000 = 13,333$ L.E.

Simulation of 20 insurance calls:

Trial	Flip	Interest	Not Interest	Roll	0	10	20	Value
1	H	X		4		X		10,000
2	H	X		4		X		10,000
3	T		X					
4	H	X		1	X			
5	H	X		2	X			
6	T		X					
7	H	X		3	X			
8	H	X		6			X	20,000
9	T		X					
10	T		X					
11	H	X		2	X			
12	T		X					
13	H	X		3	X			
14	T		X					
15	H	X		6			X	20,000
16	H	X		4		X		10,000
17	H	X		2	X			
18	H	X		4		X		10,000
19	T		X					
20	H	X		6			X	20,000

From the table, 13 of the visits will result in a “No sale”.

$P(\text{No sale}) = 13/20 = 0.65$

Given Sale, we have $4 \times 10,000$ and $3 \times 20,000$ policies sold, so:

$P(10000) = 4/7 \approx 0.5714$

$P(20000) = 3/7 \approx 0.4286$

The Expected Value = $0.5714 \times 10000 + 0.4286 \times 20000 = 14,286$ L.E.

Comparison between Theoretical & Simulation (Validation):

	Prob. of No Sale	Prob. of Sale	Prob. of 10,000	Prob. of 20,000	Expected Value
Simulation	0.65	0.35	0.5714	0.4286	14,285
Theoretical	0.75	0.25	0.667	0.3333	13,333

Time Management Methods

There are two time references used in simulation:

1. Simulation Time:

The period of time simulated by the model – 30 min., 5 years, 100 years, whatever interval the researcher is interested in. This time is reset to zero at the beginning of the simulation run and acts as a counter to the number of simulation time units.

2. Run Time:

The time that the computer takes to simulate the period of interest.

We have two ways for time changes (Periodic & Event Scan):

The Periodic Scan or Fixed Time Increment Technique:

In this technique, the simulation clock is adjusted by one predetermined uniform unit and then examines the system to determine whether any events occurred during that interval. If any occurred, the event or events are simulated; otherwise, no action is taken.

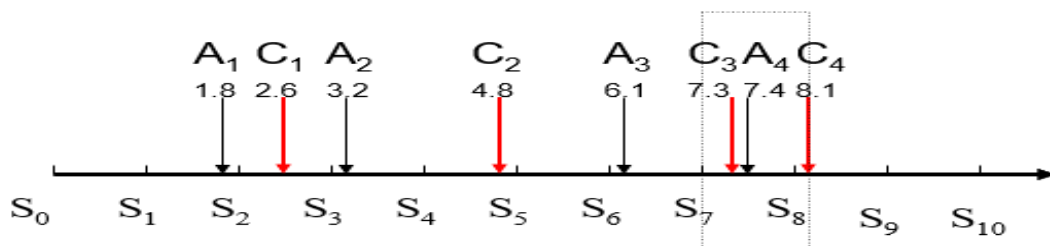
The simulation clock is then advanced another unit, and the process is repeated. In this technique, the exact time of the occurrence of a particular event is largely ignored. All events that occur during a given interval are treated as if these events occurred at the end of that interval.

Example:

Consider the simulation of a single-bay service station for a period of ten minutes. Assume that the system is empty at the beginning and at the end of the period. Assume also that four customers are serviced; their arrivals occur at simulation times 1.8, 3.2, 6.1, and 7.4, and that service completions occur at simulation times 2.6, 4.8, 7.3, and 8.1.

If a time interval of one minute is used, arrival events are denoted by A_1 , A_2 , A_3 , and A_4 , while completion events are denoted by C_1 , C_2 , C_3 , and C_4 (see the following figure).

C_3 and A_4 are both considered to have occurred at clock time 8.0; although C_3 actually occurred before A_4 .



The problem of periodic scan is that events separated in time appear to occur simultaneously. In this example, at time period $S_7 - S_8$, the system appears to be idle while actually it is not.

The big problem is to determine the length of the interval to be used. The time increment used in the periodic scan approach should be as small as possible to minimize the possibility of lost information. But the smaller the time unit, the longer the number of calculations necessary to complete the simulation.

Thus, the actual run time of the simulation and hence the cost of simulation is increased. So there is a trade-off between precision and maximum run time.

The Event Scan

In this technique, the clock is advanced by the amount necessary to trigger the occurrence of the next event. Thus, the time advance intervals are of variable lengths.

While some information can be lost in the periodic scan, the event scan avoids this problem because the clock is advanced only to the next occurrence time.

In periodic scan, the exact occurrence time of events is not known because each event is treated as if it occurred at the end of the time interval in which it occurred. This problem is overcome in the Event Scan approach.

Example

A bank of machines in a manufacturing shop breaks down according to the following interarrival time distribution. The time it takes one repair person to complete the repair of a machine is given in the service time distribution.

Interarrival Time (hours)	P(X)	Mapping of R.N.	Service Time (hours)	P(X)	Mapping of R.N.
0.5	0.30	00 – 29	0.5	0.25	00 – 24
1.0	0.22	30 – 51	1.0	0.20	25 – 44
1.5	0.16	52 – 67	2.0	0.25	45 – 69
2.0	0.10	68 – 77	3.0	0.15	70 – 84
3.0	0.14	78 – 91	4.0	0.10	85 – 94
4.0	0.08	92 – 99	5.0	0.05	95 – 99
	1			1	

Simulate the breakdown of 5 machines. Calculate the average machine downtime using two repair persons and the following random numbers sequence. Both repair persons cannot work on the same machine.

R.N.: 30, 81, 02, 91, 51, 08, 28, 44, 86, 84, 29, 08, 37, 34, 99.

Solution

RN	Interarrival Time	Time of Arrival	RN	Service Time	Repair Man1		Repair Man2		Waiting Time	Down Time
					On	Off	On	Off		
30	1.0	1.0	81	3	1	4			0	3
02	0.5	1.5	91	4			1.5	5.5	0	4
51	1.0	2.5	08	0.5	4	4.5			1.5	2
28	0.5	3.0	44	1	4.5	5.5			1.5	2.5
86	3.0	6.0	84	3	6	9			0	3
Total Down Time										14.5
Average Down Time = $14.5 / 5 = 2.9$ hours.										

Example for a Continuous Distribution:

Simulate a system M/M/1 with arrival rate $\lambda = 20/h$ and service rate $\mu = 25/h$. Assuming the system is empty at time zero.

Using:

1. Time Scan (Periodic Scan)
2. Event Scan

Solution:

1. Using Time Scan:

$$P(\text{arrival in one minute period}) = 1 - e^{(-\lambda t)} = 1 - e^{(-1/3 * 1)} \approx 0.281$$

$$P(\text{departure in one minute period}) = 1 - e^{(-\mu t)} = 1 - e^{(-5/12 * 1)} \approx 0.341$$

So let:

- $0 \leq R.N. \leq 0.281 \rightarrow \text{Arrival}$
- $0 \leq R.N. \leq 0.341 \rightarrow \text{Departure}$

Time in min.	Customer in System	Random Number Arrival	Random Number Departure	Arrival Yes/No	Departure Yes/No
0	0	0.096	-	Yes	-
1	1	0.569	0.665	No	No
2	1	0.764	0.842	No	No
3	1	0.492	0.224	No	Yes
4	0	0.950	-	No	-
5	0	0.610	-	No	-
6	0	0.145	-	Yes	-
7	1	0.484	0.552	No	No
8	1	0.350	0.590	No	No
9	1	0.450	0.410	No	Yes
10	0	0.802	-	No	-

2. Using Event Scan (Event Driven):

The time until the next arrival is given by:

$$X_{arr.} = \frac{-\ln(1-R)}{\lambda} \quad \lambda = \frac{1}{3} \text{ min.}$$

$$= -3\ln(1-R)$$

The time until the next departure is given by: $t_{dep} = -(1/\mu) \ln(1-R)$

$$X_{dep.} = \frac{-12\ln(1-R)}{5} \quad \mu = \frac{5}{12} \text{ min.}$$

Time in min.	Event	Customer in System	Random Number Arrival	Random Number Departure	Min. until next Arr.	Min. until next Dep.	Next Event
0	-	0	0.096	-	0.303	-	A
0.303	A	1	0.569	0.665	2.525	2.625	A
2.828	A	2	0.764	-	4.332	0.1	C
2.928	C	1	-	0.842	4.232	4.428	A
7.160	A	2	0.492	-	2.032	0.196	C
7.356	C	1	-	0.224	1.836	0.609	C

Example for Discrete Distribution:

Two-stage assembly lines are producing a large product. There is no room in between the two lines; so the output of the first line is fed to the second line directly, else the first line must wait until the second line is free. The service distributions for both lines are given below:

Service seconds	Time Freq. for 1 st line	R.N. Allocation	Time Freq. for 2nd line	R.N Allocation
10	4	00 – 03	4	00 - 07
20	6	04 - 09	5	08 - 17
30	10	10 – 19	6	18 - 29
40	20	20 – 39	7	30 - 43
50	40	40 – 79	10	44 - 63
60	11	80 – 90	8	64 - 79
70	5	91 – 95	6	80 - 91
80	4	96 - 99	4	92 - 99
	100		50	

Total service time for 1st line = 470 seconds

Total service time for 2nd line = 430 seconds

Item	First Line				Waiting Time	Second Line			
	No.	R.N.	Start Time	Service Time		Finish Time	R.N.	Start Time	Service Time
1	56	00	50	50	20	83	50	70	120
2	55	50	50	100		47	120	50	170
3	84	120	60	180		08	180	20	200
4	36	180	40	220		05	220	10	230
5	26	220	40	260		42	260	40	300
6	95	260	70	330	30	95	330	80	410
7	66	330	50	380		17	410	20	430
8	03	410	10	420		21	430	30	460
9	57	430	50	480		31	480	40	520
10	69	480	50	530		90	530	70	600
470 ← 60									
•The output rate average = $600/10 = 60$ seconds •Utilization of 1st line = $470/530 = 88.7\%$ •Utilization of <u>1nd</u> line = $430/550 = 78.2\%$									

Output rate average = $600 / 10 = 60$ seconds

Utilization of 1st line = $470 / 530 \approx 88.7\%$

Utilization of 2nd line = $430 / 550 \approx 78.2\%$